01/30 F ①

1. ${}_nP_r = \dfrac{n!}{(n-r)!}$

n distinct elements. select r out of n.

* we care about the order.

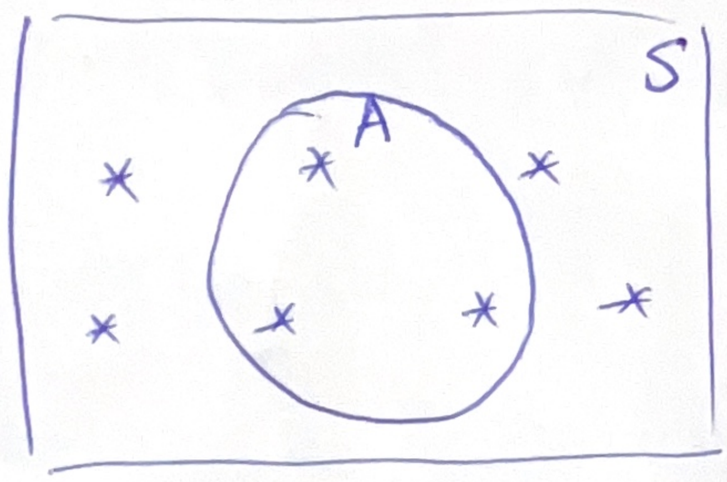
$\binom{n}{r} = \dfrac{n!}{(n-r)!\,r!}$

n distict elements. select r out of n.

* we don't care about the order.

2. Probability.

①



i> S is discrete.

each outcome in S is equally likely to happen.

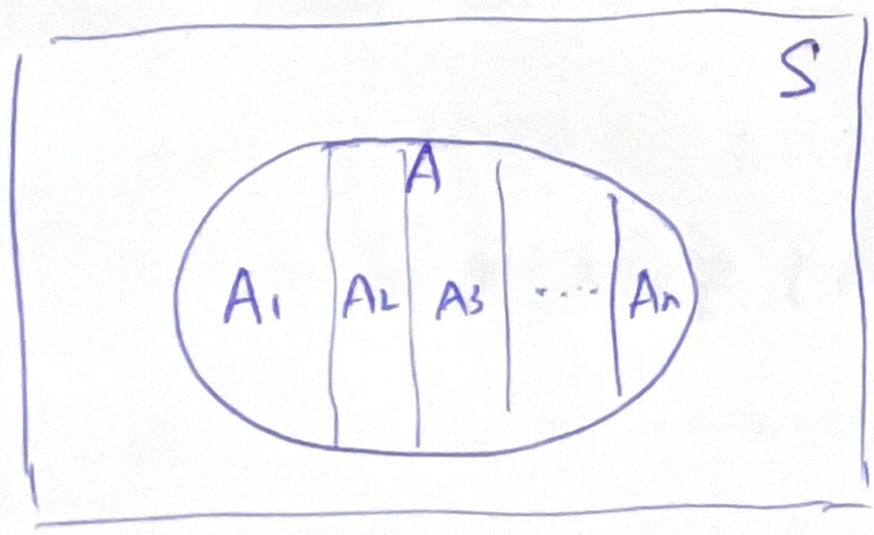
$P(A) = \dfrac{\# A}{\# S}$

Example: i> $S = \{1, 2, 3, 4, 5, 6\}$

ii>. $S = \{$total # of options to deal 10 cards$\}$

$A = \{$exact 5 hearts$\}$

②



i> S is discrete

ii>. we don't know # A, but we know the probabilities of the components comprising A.

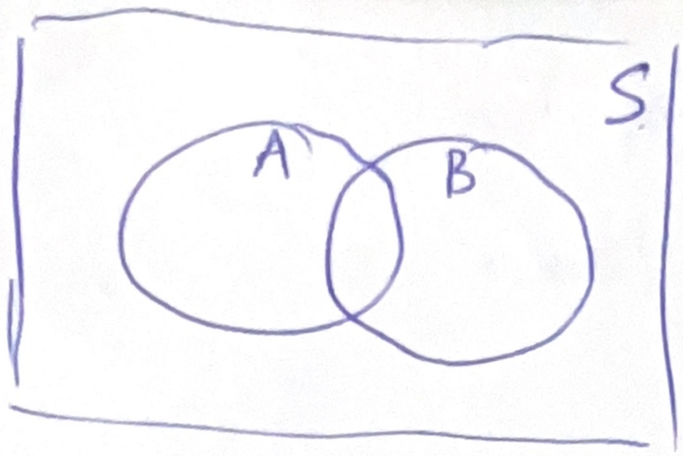
m.e. between components

$P(A) = P(A_1) + P(A_2) + \cdots + P(A_n)$

Example: Grades for STAT 2020 Exam 1 in Fall 2025.

	< 50	5%
[50, 60)	50 - 59	10%
↑	60 - 69	15%
	⋮	⋮
↓	90-100	5%

3



i> No requirements for S.

S could be either discrete or continuous

ii> Additive Rules

general rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

4. Independence.

A and B are independent.

$P(A \cap B) \overset{ind}{=} P(A) \cdot P(B)$

Conditional Probability

Def'n: the event A occurs given that event B has occurred is.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{provided } P(B) \neq 0.$$

Theorem: general product rule. [no independence required]

For any events A and B.

$$\boxed{\begin{aligned} P(A \cap B) &= P(A)\,P(B|A) \\ &= P(B)\,P(A|B). \end{aligned}}$$